Vallabhaneni Yaswanth Krishna

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SUMMARY

Results-driven B.Tech Computer Science undergraduate with hands-on experience in **machine learning** and **intelligent system deployment**. Proven ability to build and deploy **high-accuracy ML models (up to 99%)**. Strong foundation in **DSA, OOP, and data-driven problem solving**, with a demonstrated track record of converting ideas into **scalable, production-ready systems**.

EDUCATION

Vellore Institute of Technology – Andhra Pradesh — Expected May 2027
B.Tech, Computer Science and Engineering | CGPA: 9.02 — *India*

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming, Artificial Intelligence, Applications of Artificial Intelligence, Machine Learning, Natural Language Processing, Software Engineering, Deep Learning, Reinforcement Learning

EXPERIENCE

Technex Cloud Private Limited | ***AIML Engineer Intern*** — Jul 2025 – Jan 2026
India

- Trained and evaluated ML models using Python and scikit-learn, achieving approximately 85–90% validation accuracy.
- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) on structured datasets.
- Worked on end-to-end machine learning model development and deployment for data-driven educational applications.

SKILLS

Languages: C/C++, Java, Python, SQL
Technologies & Tools: AWS, EC2, Docker
Core Competencies: DSA, DBMS, DAA

PROJECTS

PulseVera – News Website
Full-Stack Web Development | Live: pulsevera-newswebsite.vercel.app

- Designed and deployed PulseVera, a live news website hosted on Vercel.
- Built and shipped the project end-to-end, covering front-end UI, deployment pipeline, and hosting configuration.

Diagnosis Prediction using Machine Learning and Deep Learning — *Aug 2025 – Nov 2025*
Python, ANN, CNN, RNN

- Developed a machine learning-based healthcare assistance system for early diagnosis prediction.
- Achieved **99% prediction accuracy** using an Artificial Neural Network (ANN).
- Benchmarked results against CNN and RNN architectures to validate model selection.

Fake News Detection and Explainable Misinformation Analysis using NLP and LLMs — *Aug 2025 – Jan 2026*
Python, BERT

- Built an NLP pipeline to detect fake news on social media and classify content as fact or fabricated.
- Integrated an LLM-based explanation layer (e.g., ChatGPT) to generate human-readable justifications for each classification.
- Achieved approximately **92% accuracy**.

Walkie-Talkie using the ESP32 Microcontroller — *May 2025 – Jul 2025*

- Built a radio-wave-based communication device to assist workers in mines and other critical, low-visibility situations.
- Implemented text-message relay between users for scenarios where voice communication is not audible, along with real-time temperature sensing at the device's location.

Traffic Light Optimization with Reinforcement Learning
Python, TensorFlow/Keras, SUMO, TraCI API, Deep Q-Learning

- Designed an autonomous traffic signal control system using a Deep Q-Network (DQN), integrating the SUMO traffic simulator with a Python RL agent through the TraCI interface.
- Modeled the intersection state as an 80-dimensional binary occupancy vector across four approach lanes, and generated realistic bursty traffic using a Weibull arrival distribution.
- Built a DQN with an 80-neuron input layer, two hidden layers (400 and 200 neurons, ReLU), and a 4-neuron output layer, trained with experience replay and an epsilon-greedy policy over 500 episodes.
- Reduced average queue length by **62.5%** (from ~1.6 to ~0.6 vehicles) compared to a traditional fixed-time signal controller, validated through a Bellman-equation-based reward formulation.

Intelligent Navigation Framework for Energy-Constrained Autonomous Mobile Robots
Python, A Search, Energy-Aware Path Planning*

- Developed a lightweight energy-aware path planning framework (LEAP) for warehouse Autonomous Mobile Robots (AMRs), extending classical A* search with a cost function combining terrain difficulty, payload weight, and battery constraints.
- Modeled the warehouse as a 20x20 grid containing obstacles, rough-terrain regions, and charging stations, with battery-feasibility checks to guarantee mission completion across multiple goals.
- Generated a synthetic multi-map dataset with varying obstacle densities and terrain distributions to benchmark the planner under diverse navigation scenarios.
- Achieved lower energy consumption and fewer charging stops than baseline shortest-path methods, with only a marginal increase in path length across all tested maps.

CERTIFICATIONS

- Completed Generative AI Certification, gaining hands-on exposure to large language models and AI-driven applications.
- Earned HackerRank 5-Star Certification in SQL.
- AWS Global Certifications.
- MongoDB Certification on DBA.

ACHIEVEMENTS

- Solved 150+ problems on LeetCode and GeeksforGeeks.
- Solved the complete SQL 50 problem set on LeetCode.
- Achieved 50 and 100 Days LeetCode Badges.
- Participated in Tata Hackathon Code Vita.